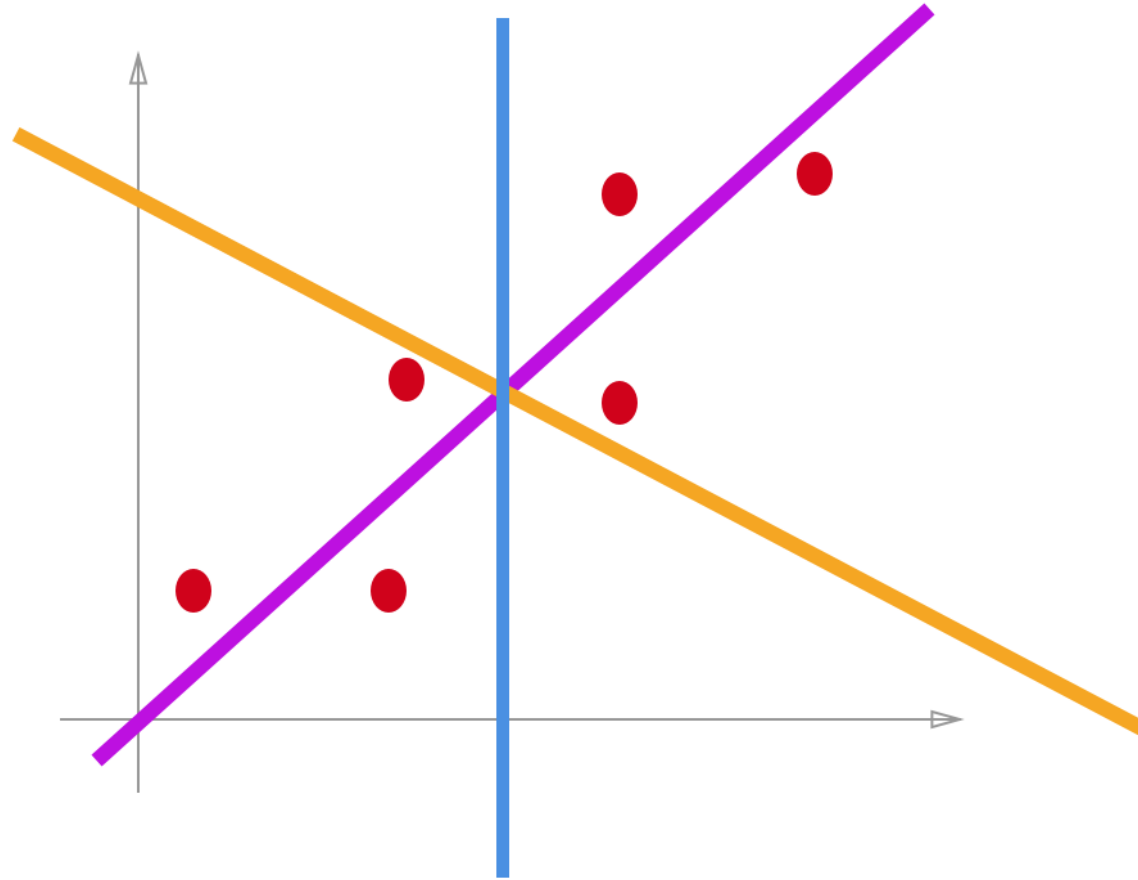
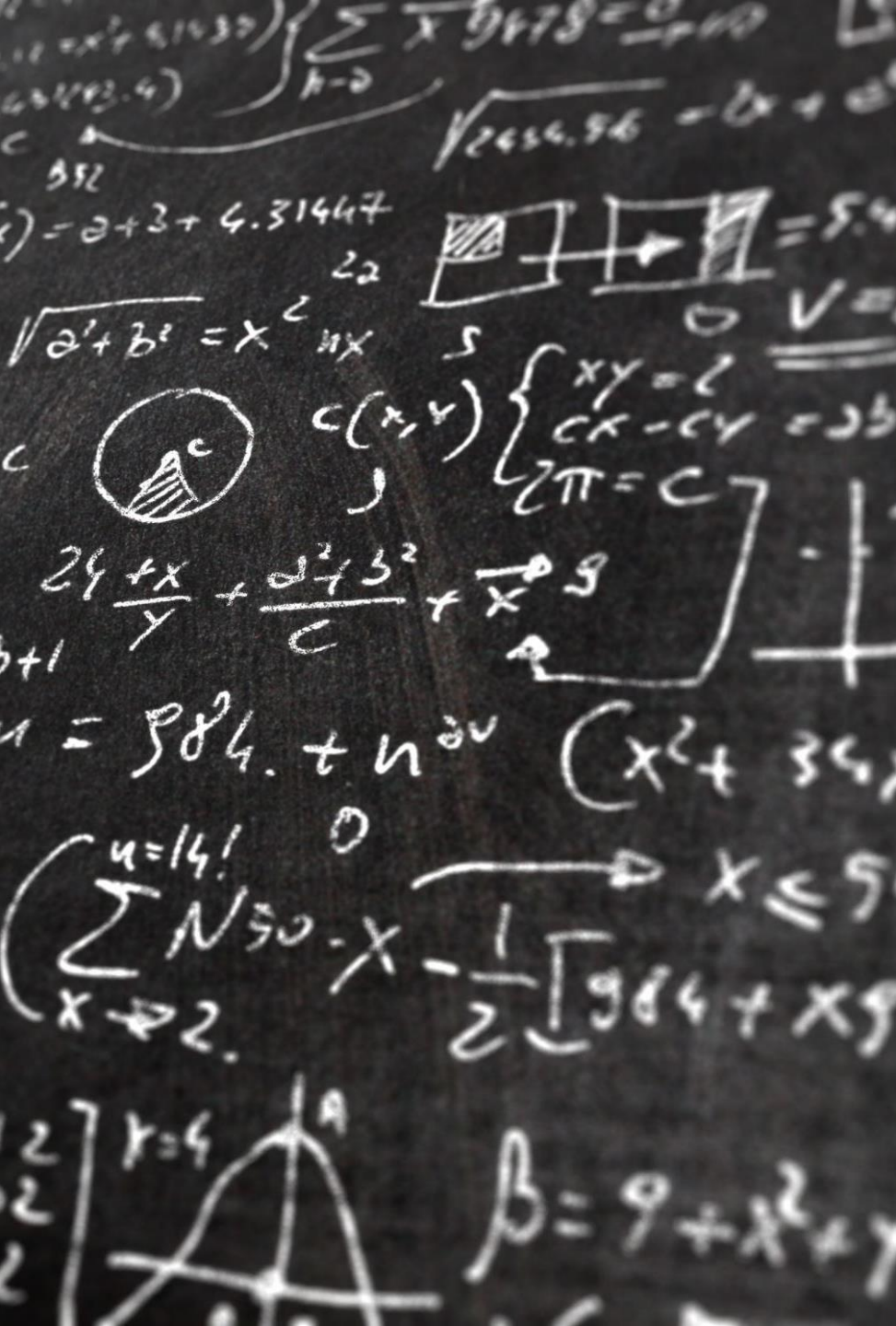


Principal Component Analysis (PCA)





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- This lecture introduces eigenvectors and their relationship to matrices in plain language and without a great deal of math.
 - It builds on those ideas to explain covariance and principal component analysis.
 - The tutorial has been take from the following publication:

Smith, L. I. (2002). *A tutorial on principal components analysis*.

Standard Deviation

- Statisticians are usually concerned with taking a *sample* of a *population*.
- To use election polls as an example, the population is all the people in the country, whereas a sample is a subset of the population that the statisticians measure.



Standard Deviation

The great thing about statistics is that by only measuring (in this case by doing a phone survey or similar) a sample of the population, you can work out what is most likely to be the measurement if you used the entire population.



Standard Deviation

The Standard Deviation (S) of a data set is a measure of how spread out the data is.

Assume X is the dataset, $\bar{x}$

Is the mean, n are the number of elements in the dataset

$$s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{(n-1)}}$$

Standard Deviation

For example, these two data sets have exactly the same mean (10), but are obviously quite different:

[0 8 12 20] and [8 9 11 12]



Set 1:

X	$(X - \bar{X})$	$(X - \bar{X})^2$
0	-10	100
8	-2	4
12	2	4
20	10	100
Total		208
Divided by (n-1)		69.333
Square Root		8.3266

Set 2:

X_i	$(X_i - \bar{X})$	$(X_i - \bar{X})^2$
8	-2	4
9	-1	1
11	1	1
12	2	4
Total		10
Divided by (n-1)		3.333
Square Root		1.8257

COVARIANCE

- Many data sets have more than one dimension, and the aim of the statistical analysis of these data sets is usually to see if there is any relationship between the dimensions.



COVARIANCE

- Covariance is a measure of how much the dimensions vary from the mean *with respect to each other*.
- Covariance is always measured between 2 dimensions.
- If you calculate the covariance between one dimension and itself, you get the variance.



COVARIANCE

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{(n - 1)}$$

$$C = \begin{pmatrix} cov(x, x) & cov(x, y) & cov(x, z) \\ cov(y, x) & cov(y, y) & cov(y, z) \\ cov(z, x) & cov(z, y) & cov(z, z) \end{pmatrix}$$

The covariance
Matrix

For example, from a 3 dimensional data set (dimensions x , y , z) you could calculate

Linear Transformations

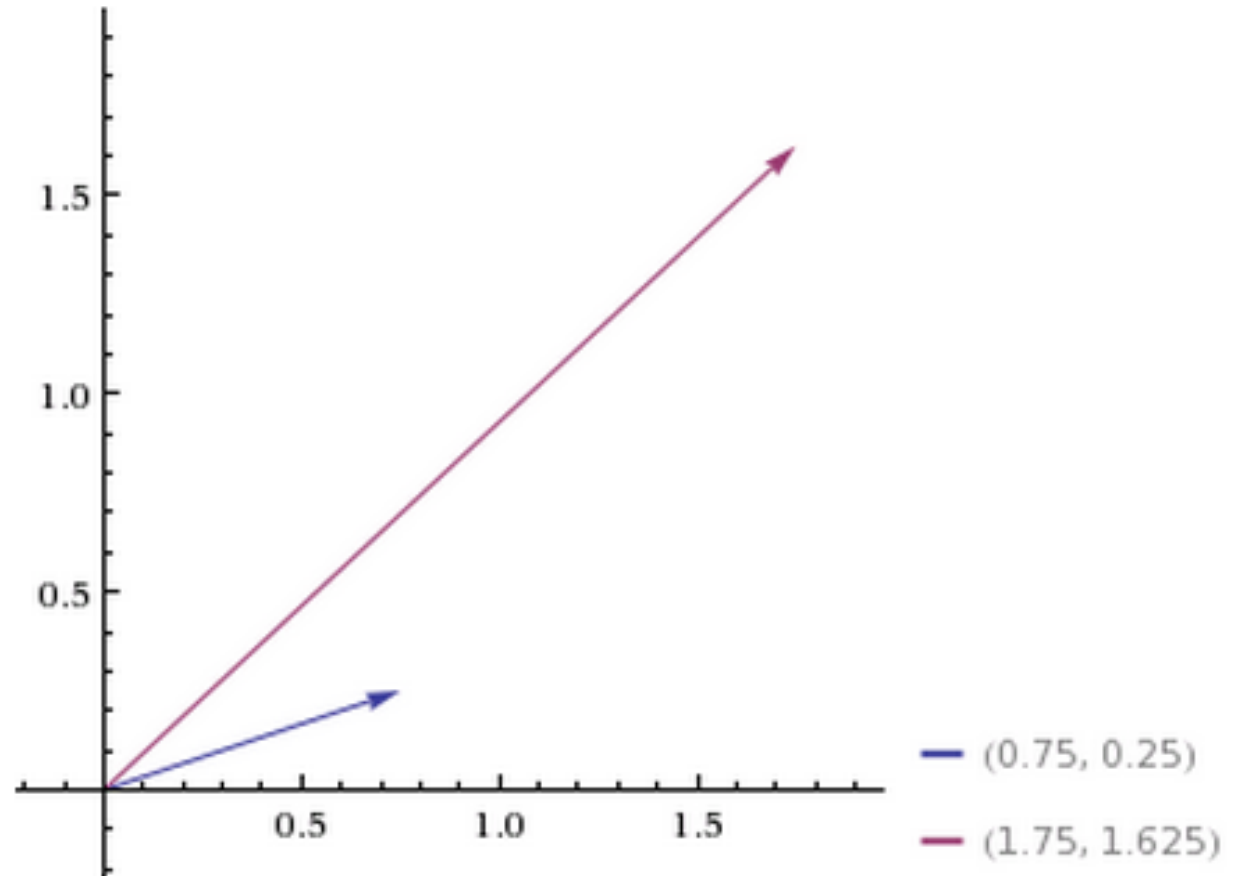
- Matrices are useful because you can do things with them like add and multiply. If you multiply a vector v by a matrix A , you get another vector b , and you could say that the matrix performed a linear transformation on the input vector.
- $Av = b$
- It maps one vector v to another, b .

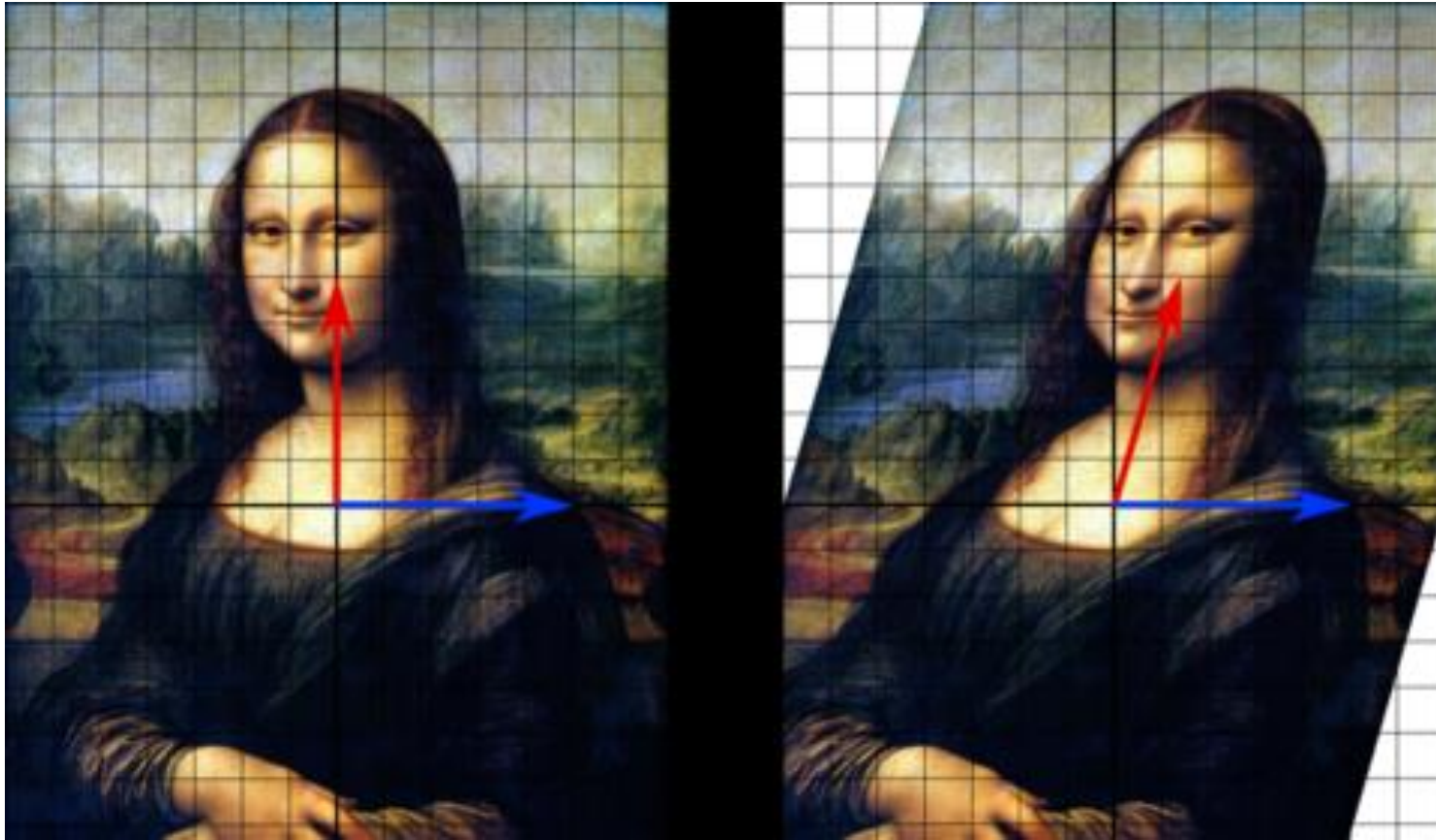
$$\begin{matrix} A & v & b \\ \begin{bmatrix} 2 & 1 \\ 1.5 & 2 \end{bmatrix} * \begin{matrix} 0.75 \\ 0.25 \end{matrix} = & \begin{matrix} 1.75 \\ 1.625 \end{matrix} \end{matrix}$$

Linear Transformations

So A turned v into b . In the graph below, we see how the matrix mapped the short, low line v , to the long, high one, b .

Vector plot:





EIGENVECTOR

You can imagine a matrix like a gust of wind, an invisible force that produces a visible result. And a gust of wind must blow in a certain direction. The eigenvector tells you the direction the matrix is blowing in.

(Credit: Wikipedia)

What properties do these eigenvectors have?

- You should first know that eigenvectors can only be found for square matrices. And, not every square matrix has eigenvectors.
- And, given an $n \times n$ matrix that does have eigenvectors, there are n of them. Given 3×3 matrix, there are 3 eigenvectors.



What properties do these eigenvectors have?

$$2 \times \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix} \times \begin{pmatrix} 6 \\ 4 \end{pmatrix} = \begin{pmatrix} 24 \\ 16 \end{pmatrix} = 4 \times \begin{pmatrix} 6 \\ 4 \end{pmatrix}$$

- Another property of eigenvectors is that even if I scale the vector by some amount before I multiply it, I still get the same multiple of it as a result.
- This is because if you scale a vector by some amount, all you are doing is making it longer,

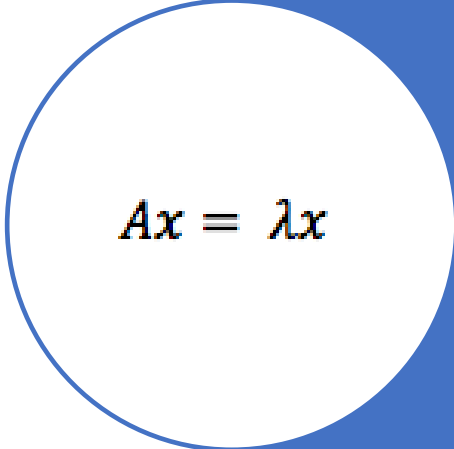
What
properties do
these
eigenvectors
have?

- All the eigenvectors of a matrix are *perpendicular* also known as ***Orthogonal***, ie. at right angles to each other, no matter how many dimensions you have.
- This is important because it means that you can express the data in terms of these perpendicular eigenvectors, instead of expressing them in terms of the x and y axes.

EIGENVECTOR

The definition of an Eigenvector, therefore, is a vector that responds to a matrix as though that matrix were a scalar coefficient. In this equation, A is the matrix, x the vector, and λ the scalar coefficient,

Eigenvectors are axes along which linear transformation acts, stretching or compressing input vectors.


$$Ax = \lambda x$$

Eigenvalues

- Eigenvalues are closely related to eigenvectors.
- Notice how, in the example in slide 16, the amount by which the original vector was scaled after multiplication by the square matrix was the same.
- In that example, the value was 4. 4 is the eigenvalue associated with that eigenvector.

Principal Component Analysis (PCA)

- Principal components are a new coordinate system
- Given data on D variables, the hope is that the data points will lie mainly in a linear subspace of dimension lower than D .

Principal Component Analysis (PCA)

- In practice, the data will usually not lie exactly in some lower-dimensional subspace but we might be able to find such a reduced subspace of dimension $d < D$ which maintains most of the variability in the data.
- There will be d new variables defining the subspace.

Principal Component Analysis (PCA)

- The new variables, which form a new coordinate system are called principal components (PCs) and we will denote them by $u_1, \dots, u_D$

Principal Component Analysis (PCA)

- They are orthogonal linear transformations of the original variables, so there are at most D of them.
- Hopefully, we will need only $d \ll D$ PCs, $u_1, \dots, u_d$, to approximate the space spanned by the values of $x_1, \dots, x_D$.

Principal Component Analysis (PCA)

- Specifically, let PC_j be a linear combination of $x_1, \dots, x_D$

defined by coefficients or weights $\mathbf{w}^{(j)} = \begin{pmatrix} w_1^{(j)} \\ \vdots \\ w_D^{(j)} \end{pmatrix}$

i.e. $u_j = w_1^{(j)}x_1 + \dots + w_D^{(j)}x_D = \mathbf{w}^{(j)T} \mathbf{x}$

Principal Component Analysis (PCA)

- We chose the first PC, u_1 to have maximum variance (so capturing as much of the variability in $x_1, \dots, x_D$ as possible).
- Subsequent PC will take up successively smaller parts of the total variability.

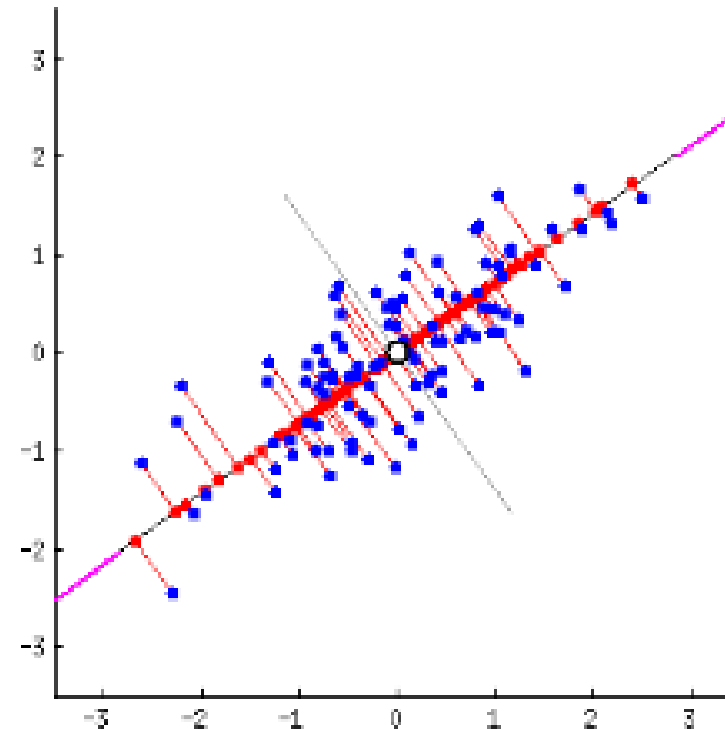
Principal Component Analysis (PCA)

- PCA attempts to draw straight, explanatory lines through data, like linear regression.
- Each straight line represents a “principal component,” or a relationship between an independent and dependent variable.

Principal Component Analysis (PCA)

- While there are as many principal components as there are dimensions in the data, PCA's role is to prioritize them.
- The first principal component bisects a scatterplot with a straight line in a way that explains the most variance; that is, it follows the longest dimension of the data.
- The second principal component cuts through the data perpendicular to the first, fitting the errors produced by the first.

There are only two principal components in the graph, but if it were three-dimensional, the third component would fit the errors from the first and second principal components, and so forth.



Principal Component Analysis (PCA) Usage

- A more common way of speeding up a learning algorithm is by using Principal Component Analysis (PCA).
- If your learning algorithm is too slow because the input dimension is too high, then using PCA to speed it up can be a reasonable choice. This is probably the most common application of PCA.
- Another common application of PCA is for data visualization.